

Independent Reproduction and Convergence Analysis of the Connes–Consani–Moscovici Zeta Spectral Triple

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Abstract

We independently implement the Connes–Consani–Moscovici (CCM) operator construction [1] in Rust at arbitrary precision (MPFR/GMP) and extend the first-zero comparison from 55 to **1495.14 measured matching decimal digits** at $(\lambda^2=1200, N=970)$ with 6708 working bits. At $(\lambda^2=1000, N=800)$, increasing requested precision from HP-1000 to HP-2000 improves agreement from 1019.0 to 1260.36 digits, separating arithmetic limitation from the resolved finite-configuration error. We characterize the construction’s convergence empirically: accuracy follows a smoothly saturating basis-size ramp that is approximately linear over the observed early range, then approaches a first-zero plateau that tracks the Weil eigenvalue depth $-\log_{10} |\varepsilon_N|$ with a slowly varying conversion gap. Along the sampled cutoff-and-basis paths, $|\varepsilon_N|$ decreases by hundreds of decimal orders over the listed prime-count doublings, although the finite data do not determine an asymptotic decay class. Across the sampled fixed- N cutoff sweep, first-zero accuracy grows monotonically with λ and closely tracks the eigenvalue depth $-\log_{10} |\varepsilon_N|$; no monotonicity claim is made outside that range. The unrestricted lowest state is essentially even at every tested above-floor configuration. At the five headline parity configurations, positive even–odd and internal even–sector gaps additionally support numerical simplicity. Analytic interval assembly and two independent inertia calculations certify positive, simple, even ground states at $\lambda^2 = 13, N = 10$ and $N = 120$. Propagating the analytic eigenstate error also certifies the finite-source root comparisons at these two configurations, including the 55-digit reproduction. A residual-to-gap bound explains the observed parity noise. A 21×21 matrix from six primes already yields 21.585 matching digits; an exact reference-free census of its serialized secular source verifies the root count and spectral indexing. All results are reproducible from the accompanying source-available implementation. This work has been shared with the CCM authors and reviewed by the lead author, Professor Alain Connes.

1 Introduction

The Riemann Hypothesis (RH), asserting that all nontrivial zeros of the Riemann zeta function $\zeta(s)$ satisfy $\Re(s) = 1/2$, remains one of the central open problems in mathematics. Among approaches toward its resolution, the Hilbert–Pólya conjecture — positing the existence of a self-adjoint operator whose spectrum encodes the imaginary parts of the nontrivial zeros — has motivated extensive work in spectral theory and noncommutative geometry.

In November 2025, Connes, Consani, and Moscovici [1] proposed the perturbed scaling-operator family $D_{\log}^{(\lambda, N)}$, determined by a finite Weil-form restriction. Its spectra exhibit striking numerical agreement with the nontrivial zeros of $\zeta(1/2 + is)$ in the tested finite configurations. Their construction uses the Weil quadratic form built from the explicit formula, restricted to

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primes $p \leq \lambda^2$, and produces a $(2N+1) \times (2N+1)$ self-adjoint matrix whose smallest eigenvalue ε_N is observed to decrease over the tested families as λ and N increase. They reported accuracy of 2.5×10^{-55} on the first zero at $(\lambda^2=13, N=120)$ with 200-digit working precision. Subsequent independent work by Groskin reproduced the high-precision zero approximation from the truncated Weil form across multiple cutoffs [2].

Recent continuum results provide complementary context for the parity observation below. Suzuki proves that, for sufficiently small compact-window half-width $a > 0$, the localized Weil operator has a positive, simple lowest eigenvalue with an even eigenfunction [3]. Zhu reports an unconditional certificate at $a = 0.8$, equivalently $\lambda^2 = e^{2a} \approx 4.95$, including a simple even continuum ground state [4]. The finite reduction, tail estimates, and two-sector separation in that work provide a constructive route from computed gaps to a continuum certificate. The numerical full-space tests reported here begin at $\lambda^2 = 13$ and extend the tested parameter range; quantitative tail and gap control is needed to connect these finite tests to larger continuum windows. Zhu also proposes a Landau–Widom-type decay law for the continuum compact-window infimum, with a fitted leading constant. That proposal concerns a different limit and is not used here as an asymptotic law for the finite Galerkin quantity $\varepsilon_N(\lambda^2, N)$ or for secular-root error.

In this paper we:

1. Independently reproduce the CCM headline result on a Rust + MPFR/GMP software stack;
2. Extend the first-zero comparison to **1495.14 measured matching digits** at $(C, N) = (1200, 970)$, where $C = \lambda^2$. The paired-precision experiment at $(C, N) = (1000, 800)$ distinguishes arithmetic limitation from finite-configuration error;
3. Systematically characterize the convergence behavior as a function of both N (basis size) and λ (prime cutoff);
4. Test the simple-even ground-state condition using unrestricted eigenvectors and the low spectra of both parity sectors. We give a residual-to-gap estimate for parity error and an exact reference-free root census for the small-matrix example.

All computations are reproducible from the source-available implementation at <https://github.com/TeamXcelerator/ccm-reproduction-and-convergence>.

2 Mathematical Setup

We briefly recall the CCM construction, following the notation of [1].

2.1 The Weil Quadratic Form

Let $\lambda > 1$, $C = \lambda^2$, and $L = 2 \ln \lambda = \ln C$. The Hilbert space is $\mathcal{H}_\lambda = L^2([\lambda^{-1}, \lambda], du/u)$. The finite Fourier subspace $E_N \subset \mathcal{H}_\lambda$ has orthonormal basis

$$V_n(u) = \frac{1}{\sqrt{L}} e^{2\pi i n \ln(\lambda u)/L}, \quad n \in \{-N, \dots, N\}.$$

The restriction of the Weil quadratic form QW_λ to E_N is represented by the real symmetric $(2N+1) \times (2N+1)$ matrix τ with entries

$$\tau_{n,m} = W_{0,2}(V_n, V_m) - W_R(V_n, V_m) - \sum_{\substack{k=p^j \\ k \leq \lambda^2}} \frac{\log p}{\sqrt{k}} q(V_n, V_m)(\log k),$$

where $W_{0,2}$ is a closed-form rank-two pole correction, W_R is the archimedean (gamma-factor) contribution computed via Gauss–Legendre quadrature, and the prime-power sum encodes the von Mangoldt function $\Lambda(k) = \log p$ for $k = p^j$.

2.2 From Eigenvector to Spectral Ordinates

Let ε_N denote the smallest eigenvalue of τ and $\xi = (\xi_{-N}, \dots, \xi_N)$ the corresponding eigenvector. In the reproduction pipeline we analyze the even representative ($\xi_{-j} = \xi_j$) and normalize it so that $\sum_j \xi_j = \sqrt{L}$. Define the secular rational function

$$R(s) = \xi_0 + 2s \sum_{j=1}^N \frac{\xi_j}{s - j^2}.$$

Let $s_1 < \dots < s_M$ denote its positive real roots, indexed in increasing order. The corresponding positive spectral ordinates are

$$\nu_r = \frac{2\pi}{L} \sqrt{s_r}.$$

The root ordinal r is not the pole-interval index j : many intervals $(j^2, (j+1)^2)$ can be empty, and at small N some roots matching low zeta ordinates can lie above the largest pole N^2 . The full perturbed scaling operator agrees with the unperturbed operator on $E_N^\perp$, retaining the lattice values $2\pi j/L$ for $|j| > N$ [1, Proposition 5.7]. We therefore distinguish the movable-root rank r , a reference-zero label k , and the ordinal in the full positive spectrum. The tables compare the selected roots with the ordered positive ordinates γ_k of $\zeta(1/2 + i\gamma_k) = 0$; identification by reference matching is separate from counting the full spectrum. Section 4.4 gives an exact count for the small example.

All tables reporting “matching digits” use the relative metric

$$D_r(\lambda^2, N, P) := -\log_{10} \left(\frac{|\nu_r - \gamma_r|}{|\gamma_r|} \right). \quad (1)$$

Section 4.1 also reports absolute error to compare directly with the original CCM headline result.

3 Implementation

Our implementation uses the Xcelerator Toolkit,¹ a publicly available library of high-precision spectral methods for analytic number theory, pinned to a tagged release for reproducibility. The toolkit provides the core numerical primitives (Gauss–Legendre quadrature with disk-cached HP nodes, LU-based inverse iteration, Newton refinement) and the CCM-specific construction (Weil-form matrix assembly, eigenvector extraction, rational-function root-finding). The paper-specific repository contains the CLI harness, reproduction scripts, and independent verification supplements. The production numerical implementation is shared via the toolkit.

The toolkit is built on:

- **Rust** (edition 2021) for the main codebase;
- **rug** (v1.30, wrapping MPFR/GMP) for arbitrary-precision floating-point arithmetic;
- **nalgebra** (v0.33.3) for the f64 fast-path eigendecomposition (used only by the optional `--f64-only` mode for smoke tests; the HP publication path uses the toolkit’s own HP symmetric eigendecomposition in `xc-numeric::eigen`, verified at HP-1000 against a PARI/GP 2000-digit reference [5]);

¹<https://github.com/TeamXcelerator/xcelerator-toolkit>

- **rayon** (v1.12.0 in the locked dependency graph) for parallel matrix assembly, Householder reduction, LU factorization, and eigenvalue-per-seed refinement.

Key algorithmic choices:

1. **Parity-aware eigenstate solve:** Paper claims use the direct reduced even-sector matrix. The default automatic solver first reuses an exact compatible eigenstate when available; on a cache miss it applies a retained-factor shift-invert Krylov/Rayleigh–Ritz method, always initialized from its canonical state. Cache-derived or explicitly supplied continuation states are not admitted to the persisted solve, so the retained payload cannot depend on which compatible artifacts happened to be present before the run. A residual-verified LU inverse-iteration route remains the conservative fallback when the targeted Ritz value is not isolated unambiguously. The unrestricted natural and conditional-projection routes use the full-space inverse-iteration implementation. Parity policy and solver route are part of artifact identity, so numerically equivalent states from different operation sequences cannot be confused in cache reuse.
2. **Halley’s method for $R(s)$ zeros:** The rational function $R(s) = \xi_0 + 2s \sum_{j=1}^N \xi_j / (s - j^2)$ has analytically available first and second derivatives, enabling Halley’s method (cubic convergence) as the default root-finder. The paper-claim scripts seed refinement from a toolkit-owned reference dataset whose zeros were independently computed to 2,500 decimal digits using rigorous Arb interval arithmetic [6]; their leading 1,000 digits were cross-validated against Odlyzko’s tabulation [7]. The reference ordinate selects which finite secular root is refined; it is not substituted into the result. An accepted iterate must satisfy the finite secular equation and the solver’s convergence tests. Independent finite-source discovery is also available for unseeded research runs. The point solver distinguishes four outcomes: **Converged** (the requested correction target was met), **Stagnated** (a finite approximation stopped improving), **Approximate** (the iteration limit was reached with a finite value), and **Failed** (no usable finite value). Computed workflows retain finite nonconverged values with diagnostics; they are not mislabeled as converged or certified.
3. **Gauss–Legendre quadrature with disk cache:** GL nodes and weights at a given quadrature order and bit precision are computed once and cached for reuse across runs. This quadrature evaluates the finite- x representation of the full archimedean matrix entries used by the CCM construction; it does not introduce the separate finite-frequency cutoff T used in Groskin’s implementation and studied there for truncation sensitivity [2].

Independent matrix assembly. At $C = 13$, $N = 10$, a separate tanh–sinh integration of the defining CCM components [1, Equations (3.14), (3.16), and (4.4)] checks all 441 entries, including the zero mode. The maximum absolute differences from the analytic interval-assembly midpoints are 4.05×10^{-60} at 60 decimal digits and 2.95×10^{-90} at 90 decimal digits. This independently checks the assembly formulas numerically; the interval certificates below supply the rigorous finite-matrix conclusions.

3.1 Reproducibility Infrastructure

The toolkit manages typed artifacts for quadrature rules, the archimedean and prime components, τ matrices, parity reductions, factorizations, Weil states, secular sources, root refinements, and validation evidence. Each artifact has a content-bound manifest whose semantic identity includes the numerical parameters, parity policy, algorithm route, precision contract, and dependency identities that can affect its value. Loads replay family-specific structural checks; incompatible or obsolete artifacts are rejected rather than silently reused.

Compatible artifacts are reused when available; otherwise, the reproduction scripts compute them locally. Caching accelerates repeated runs without changing the content-identified numerical route or the validation checks. Reproducibility therefore does not depend on a particular cached source being available.

The September 2026 v2.5 campaign assessed all 14 individual claim scripts over 53 capture journals. Numerical and publication summaries passed for every script, with the stated applicability policies. Claim 1c retains five supported diagnostics and eight predeclared exclusions. Claim 8 has a separate retrospective review of all four journals and 46 applicable diagnostics; its original incomplete natural-route receipts remain unchanged. The campaign summary and supplemental numerical evidence are included in the repository’s `docs/RESEARCH_EVIDENCE.md`.

4 Results

4.1 Reproduction of the CCM Headline

At ($\lambda^2=13$, $N=120$, 200-digit precision), our first computed spectral ordinate is

$$\nu_1 = 14.134725141734693790457251983562470270784257 \dots$$

(displayed truncated to 42 digits; the full HP-1000 result has $D_1 = 55.764$ under (1), see Section 4.3) with absolute error 2.4363×10^{-55} against the independently computed 2,500-digit reference. This matches the CCM paper’s reported 2.5×10^{-55} to within rounding. The result is unchanged at an HP-1000 requested-precision target, confirming that the observed accuracy tracks $\varepsilon_N \approx 3.48399 \times 10^{-59}$ rather than being limited by arithmetic precision.

4.2 Extension to 1019 Measured Digits

Table 1: Measured matching digits across configurations, all at an HP-1000 requested-precision target with a 64-bit internal guard reserve. At $\lambda^2=1000$ the reported agreement saturates guarded working precision; the construction’s intrinsic ceiling is not resolved by this run.

Configuration	First-zero digits	20th-zero digits	Requested precision
$\lambda^2=13$, $N=120$	55.764	25.339	HP-1000
$\lambda^2=100$, $N=500$	460.09	422.44	HP-1000
$\lambda^2=1000$, $N=800$	1019.0	1020.3	HP-1000

Here HP-1000 denotes the requested accuracy contract, not the raw MPFR allocation. The implementation converts 1000 decimal digits to 3322 requested bits and appends 64 guard bits, giving 3386 working bits, or approximately 1019.3 decimal digits. Root acceptance is tested against the 3322-bit requested target; the guard reserve protects that target from cancellation and accumulated rounding error. The matching-digits diagnostic, however, compares the final values at the full guarded precision, so a converged Halley iteration can expose agreement within the reserve: 1019.0 digits for the first zero in the latest run. The 20th-zero relative metric reaches 1020.3. The bit-to-decimal conversion describes significant arithmetic precision, not a strict upper bound on pointwise agreement: a representable result can lie unusually close to the comparison value. Division by $|\gamma_{20}|$ alone does not explain this effect, since floating-point precision is already relative. These guard-space measurements do not redefine HP-1000 as a 1019-digit requested calculation; they report the additional agreement preserved by the internal working precision.

Increasing precision reveals additional digits while arithmetic limits the root calculation. At the same ($\lambda^2=1000$, $N=800$), the HP-2000 run reaches 1260.36 matching digits, rather than

another full 1000 digits beyond the HP-1000 result. The matrix scale, spectral gap, coefficient errors, and local secular derivative determine how arithmetic error propagates to a root. Cutoff, basis size, and working precision must therefore be budgeted together for a desired root accuracy.

4.3 Convergence in N (Basis Size)

Table 2: Matching digits on the first five zeros vs. N at $\lambda^2=13$. Values agree to the displayed precision at HP-200 and HP-1000 across the entire sweep (see commentary below); the column shown is from the HP-1000 run.

N	Zero 1	Zero 2	Zero 3	Zero 4	Zero 5
10	21.585	16.335	11.856	5.1626	3.4915
20	35.795	32.261	30.011	26.504	24.870
30	44.506	41.267	39.262	36.208	34.844
40	50.294	47.152	45.225	42.304	41.015
50	53.827	50.721	48.822	45.948	44.685
60	55.299	52.203	50.312	47.451	46.195
80	55.654	52.560	50.670	47.811	46.556
100	55.735	52.641	50.751	47.892	46.637
120	55.764	52.670	50.779	47.921	46.666

The sweep shows a smoothly saturating ramp with two useful empirical regimes:

1. **Rapid pre-saturation growth** ($N = 10$ to $N \approx 60$): matching digits are approximately linear over the early part of the sweep, while the local rate $\Delta(\text{digits})/\Delta N$ decreases smoothly from ~ 1.4 at $N=10 \rightarrow 20$ to ~ 0.15 at $N=50 \rightarrow 60$. The average ~ 0.96 digits per unit N over $N \in [10, 40]$ is a local summary, not a global rate law.
2. **Near-plateau behavior** ($N > 60$): the first-zero accuracy approaches 55.764 digits by $N = 120$. At this configuration,

$$D_\varepsilon := -\log_{10} |\varepsilon_N| = 58.458,$$

so the first-zero plateau lies 2.694 digits below the eigenvalue depth. We therefore call it the root-accuracy plateau associated with ε_N , not the eigenvalue depth itself.

The precise local slope and plateau location are properties of the sampled (λ, N) regime. Here $\lambda^2 = 13$ and $N \in \{10, 20, 30, 40, 50, 60, 80, 100, 120\}$; the paper does not assert the same numerical break point at other cutoffs.

Precision comparison. At every N in this sweep, HP-200 and HP-1000 produce matching digits that agree to the displayed precision. The observed plateau is therefore intrinsic to the finite configuration rather than imposed by the arithmetic precision; Sections 4.6 and 4.7 describe the corresponding floor mechanism in other regimes.

4.4 The 21-Digit Result from a 21×21 Matrix

At $N = 10$ (matrix size 21×21), with only six primes ($\{2, 3, 5, 7, 11, 13\}$, corresponding to nine prime powers entering the Weil form), the first Riemann zero is matched to **21.585 decimal digits**. This demonstrates notable numerical compression: more than 21 matching digits from a 21×21 matrix.

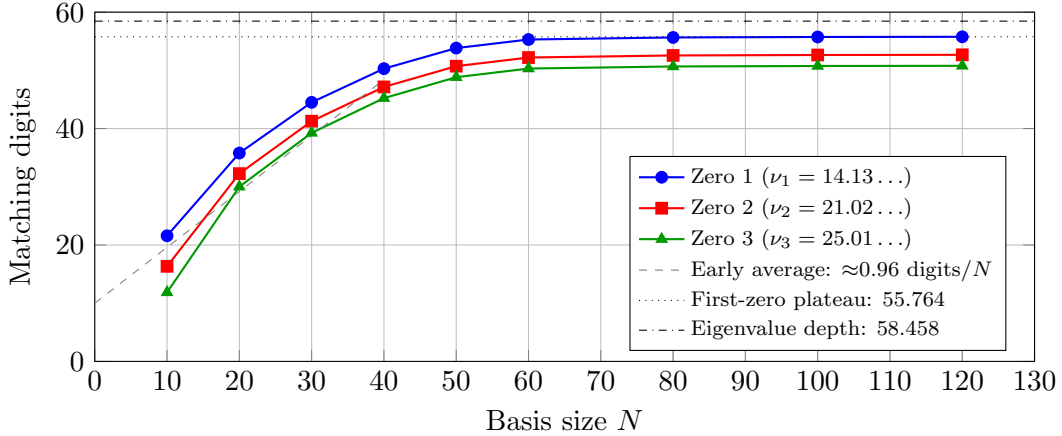


Figure 1: Convergence of matching digits vs. basis size N at $\lambda^2 = 13$, HP-1000. The two horizontal lines distinguish the first-zero plateau from the deeper Weil-eigenvalue scale.

This small configuration also illustrates that the lattice edge is not a hard root cutoff. Here

$$T_{\max} = \frac{2\pi N}{L} \approx 24.50,$$

yet the third through fifth zeta ordinates $\gamma_3, \gamma_4, \gamma_5 \approx 25.01, 30.42, 32.94$ still have accurate finite matches, and the corresponding secular roots lie above the largest pole $N^2 = 100$. An exact reference-free census now makes this example more precise. For the retained even state, interpret its serialized decimal coefficients as exact rationals and clear the poles of $R(s)$. The degree-ten numerator is squarefree, has no common factor with $\prod_{j=1}^{10} (s - j^2)$, and has exactly ten positive real roots and no negative roots. Exact Sturm isolation, independently replayed with a second rational root counter, precedes the zeta comparisons. Arb encloses the conversion to ordinates. The exact census first concerns the serialized finite source. The error transfer in Proposition 2 additionally certifies all ten root brackets for the analytic finite ground state.

Table 3: Reference-free census of the $C = 13, N = 10$ finite source. The comparison labels coincide with the first five movable ranks. Untouched scaling modes change the last two full-spectrum positions.

Movable rank / reference label	Matching digits	Full positive ordinal
1	21.585	1
2	16.335	2
3	11.856	3
4	5.163	6
5	3.491	8

For a nonlattice root, its full positive ordinal is $r + \max\{0, \lfloor \sqrt{s_r} \rfloor - N\}$ in this counted source. The third root lies above the retained Fourier edge but below the first unchanged carrier. The fourth and fifth finite matches remain valid and occupy positions six and eight. Thus the Fourier edge is not an absolute boundary for a resolved finite-root match.

4.5 Rapid Empirical Change of ε_N

These values show a rapid increase in eigenvalue depth along the listed finite-configuration path. They should not be read as a prime-count-only law: the three rows use different basis sizes and

Table 4: Selected finite-configuration values of the smallest Weil eigenvalue, with requested decimal precision and working bits. All rows use the even-sector route. The explicit N and precision columns distinguish these values from floor-limited measurements.

λ^2	N	Primes	HP / bits	ε_N	Depth
13	120	6	1000 / 3386	3.48399×10^{-59}	58.458
100	500	25	1000 / 3386	9.56481×10^{-464}	463.02
1000	800	168	2000 / 6708	$3.92215 \times 10^{-1264}$	1263.41

Table 5: Smallest Weil eigenvalue ε_N above the working-precision floor, with N scaled at the fixed ratio $N/\sqrt{\lambda^2} \approx 28$. HP-1500 for $\lambda^2 = 500$ –800 and HP-2000 for $\lambda^2 = 1000$ –1200, with 5047 and 6708 working bits, respectively. The even-sector route is used throughout. The D_1 column records the first-root matching digits retained by the v2.5 campaign. The final column gives the change in eigenvalue-depth digits, normalized per doubling of prime count and computed between consecutive rows; it is illustrative rather than a universal rate.

λ^2	N	Primes	ε_N	Depth	D_1	Depth/doubling
500	630	95	1.09117×10^{-929}	928.96	925.938	—
600	690	109	$2.22999 \times 10^{-1029}$	1028.65	1025.63	502.6
700	740	125	$9.45898 \times 10^{-1116}$	1115.02	1112.00	437.1
800	790	139	$1.35946 \times 10^{-1199}$	1198.87	1195.84	547.4
1000	890	168	$1.52095 \times 10^{-1362}$	1361.82	1358.79	596.1
1200	970	196	$6.79867 \times 10^{-1499}$	1498.17	1495.14	613.1

sit at different positions on the finite- N ramp. The next table gives a more systematic diagonal path with $N/\sqrt{\lambda^2} \approx 28$.

The depth grows rapidly throughout this sampled family. The per-doubling numbers depend on the parameterization and, critically, on the chosen basis path $N(\lambda)$. A different scaling of N produces different values. Six finite samples cannot establish whether the asymptotic decay is polynomial, exponential, or super-exponential. A rigorous two-parameter law for $\varepsilon_N(\lambda, N)$ remains open.

For the five retained even sources at $(C, N) = (13, 120)$, $(100, 500)$, $(1000, 800)$, $(1000, 890)$, and $(1200, 970)$, we additionally enclose the first secular root using Arb and compare it with a freshly computed rigorous zeta-zero enclosure. An independent replay verifies opposite endpoint signs, pole exclusion, and a derivative that excludes zero throughout each interval. The relative matching depths round to 55.76, 460.09, 1260.36, 1358.79, and 1495.14, respectively; the final depth is enclosed by $[1495.1385106984, 1495.1385106986]$. These are certified local roots of the exact serialized sources, including their finite decimal coefficients. The analytic eigenstate error transfer is completed below for $C = 13$. The four larger-cutoff comparisons retain their serialized-source scope.

4.6 Observed Monotone Convergence in λ at Fixed N

At fixed $N = 120$, first-zero accuracy grows monotonically across the sampled cutoff range and closely tracks the eigenvalue depth

$$D_\varepsilon(\lambda) := -\log_{10} |\varepsilon_N(\lambda)|.$$

The two quantities are not identical: their difference is the finite eigenvalue-to-root conversion gap.

For HP-200, the implementation converts the 200-decimal-digit request to 665 bits and adds 64 guard bits, yielding 729 working bits, or approximately 219.5 decimal digits. Root acceptance

Table 6: First-zero accuracy vs. λ^2 at $N = 120$, comparing HP-200 and HP-1000 requested precision. The final column shows the slowly varying conversion gap $C_{\text{conv}} = D_\varepsilon - D_1$.

λ^2	Primes	ε_N at HP-1000	HP-200	HP-1000	C_{conv}
13	6	3.48399×10^{-59}	55.764	55.764	2.694
20	8	2.50471×10^{-96}	92.817	92.817	2.784
30	10	2.19404×10^{-135}	131.80	131.80	2.859
50	15	6.89051×10^{-174}	170.19	170.19	2.972
100	25	3.48656×10^{-215}	211.29	211.29	3.168

is still measured against the 665-bit requested target. The 211.29-digit result at $\lambda^2 = 100$ therefore uses part of the guard reserve without changing the nominal HP-200 accuracy contract. Diagnostics closer to the floor remain precision-sensitive: HP-200 gives $\varepsilon_N = 3.48676 \times 10^{-215}$, compared with 3.48656×10^{-215} at HP-1000. For the retained complete-QR sector comparison, define $\text{GapLog} = \log_{10}(|o_0|/|e_0|)$, where e_0, o_0 are the lowest even and odd values. Its values are 4.246343 and 4.246284 at 729 and 3386 bits, respectively. The full retained inputs reproduce these diagnostics; operation-route identities distinguish their low-precision variants.

Two empirical regularities emerge:

1. **Observed monotone growth over the sampled range.** First-zero accuracy increases at every listed cutoff. No fixed- N monotonicity claim is made outside this finite sweep.
2. **ε_N -tracked root accuracy.** The first-zero digits follow D_ε with a conversion gap that grows slowly from about 2.7 to 3.2 digits across the table.

The cutoff dependence is not featureless. Each prime-power threshold changes the finite matrix path and produces a localized slope kink, precisely the event structure isolated by Groskin [8]. In sampled accuracy and eigenvalue curves these events appear as fine-scale perturbations on an otherwise smooth envelope, rather than as a separate staircase law.

For practitioners, increasing the cutoff is useful only when basis size and working precision are scaled with the desired target. The present sweep establishes monotone growth only for $13 \leq \lambda^2 \leq 100$ at $N = 120$; it does not assert that increasing λ alone is sufficient outside that range.

4.7 Critical N at Fixed Working Precision

Critical N must specify the criterion being met. Resolving a requested root, reaching a specified accuracy, and entering a regime where cutoff or numerical precision limits further improvement are distinct events. Their basis requirements depend on the cutoff, the selected root, and the working precision; an accuracy threshold also depends on the requested matching depth.

Here we report a target-accuracy threshold on a tested ladder $\mathcal{N}$, holding cutoff and precision fixed. For root index r and requested depth D_* , define

$$N_{\text{crit, tested}}(D_*; r) = \min\{N \in \mathcal{N} : D_r(N) \geq D_*\},$$

with D_r assessed only where the requested root match is resolved. If no tested configuration qualifies, the target is not reached on that ladder. This is the first successful sampled value, not a certified minimum over every integer N or a guarantee for all larger bases. A persistent threshold requires additional monotonicity or uniform error bounds.

Table 2 supplies concrete sampled thresholds at $\lambda^2 = 13$. The coarse ladder first reaches 50 matching digits at $N = 40$ and 55 digits at $N = 60$. To resolve these crossings, we tested every integer in $31 \leq N \leq 40$ and $51 \leq N \leq 60$ at both HP-200 and HP-1000 (40 additional computations). The denser ladder retains the 50-digit threshold at $N = 40$ and refines the

Table 7: Large-basis configurations at HP-1000, reporting the smallest Weil eigenvalue and first- and fifth-zero matching digits. These paired measurements show gains with increasing N ; they do not locate exact critical basis sizes.

Configuration	ε_N at HP-1000	First-zero digits	5th-zero digits
$\lambda^2=50, N=200$	5.69795×10^{-221}	217.34	207.24
$\lambda^2=50, N=250$	4.51247×10^{-239}	235.45	225.42
$\lambda^2=100, N=300$	4.68074×10^{-368}	364.37	354.03
$\lambda^2=100, N=400$	6.12702×10^{-423}	419.28	409.04

55-digit threshold to $N = 56$; Table 8 gives the adjacent pairs. Every tested value before each crossing falls below its target, and every tested value from the crossing to that bracket’s upper endpoint meets it. The maximum difference in matching depth between the two precisions is less than 3.18×10^{-160} digits.

Table 8: Adjacent-integer Critical- N crossings at $C = 13$ for the first zero. HP-200 and HP-1000 agree at all digits shown. The tested integer brackets are 31–40 and 51–60, respectively.

Target	Preceding N	D_1	First passing N	D_1
50 digits	39	49.745584	40	50.293684
55 digits	55	54.708183	56	55.011854

By contrast, the early average of approximately 0.96 digits per added mode does not justify extrapolating to 200 matching digits at $N \approx 210$: the slope decreases substantially, and the observed first-zero agreement is only 55.764 digits at $N = 120$. This finite sweep also does not establish an infinite- N accuracy ceiling.

Table 7 shows that other cutoffs and larger bases reach substantially greater depths at HP-1000. To distinguish an accuracy threshold from the onset of saturation, one must compare basis-size sweeps at sufficient precision and check precision sensitivity at fixed configurations. Extra modes alone need not meet a target when cutoff or numerical error remains limiting.

The companion convergence study [9] develops the joint basis, cutoff, and precision budget.

4.8 Even-Symmetry of the Smallest Eigenvector

The simple-even ground-state condition in the CCM program motivates two finite tests: the parity of the unrestricted lowest state and its separation from the lowest odd and next even states. Let J denote the corresponding coefficient reversal, $(J\xi)_j = \xi_{-j}$. We measure the natural evenness $\|\xi - J\xi\|/\|\xi\|$ of the unforced smallest eigenvector at working precisions sufficient to resolve ε_N above the floor:

The retained sector spectra supply the simplicity test. Let e_0, e_1 be the first two even-sector eigenvalues and o_0 the lowest odd one. All five configurations have $e_0 > 0$, $o_0 - e_0 > 0$, and $e_1 - e_0 > 0$. The odd-to-even ground-value ratio ranges from 8.77×10^3 to 4.21×10^5 . Independently retained HP Sturm brackets separate the first two even indices, and the even-state residual divided by that separation is below 2.01×10^{-533} in every row. The two sector spectra and the response record are bound to a common Tau source. Together with the unrestricted states, these measurements support both simplicity and evenness of the finite ground state.

Certified finite ground states. At $C = 13$, $N = 10$ and $N = 120$, we additionally assembled the analytic cutoff-free matrix in interval arithmetic, using 512 and 1024 bits, respectively. Arb interval LDL^T proves positive definiteness of the full matrix and encloses e_0, e_1, o_0 by shifted inertia. The parity restrictions use the exact reflection symmetry and integer bases, with

Table 9: Symmetry properties of the natural (unforced) smallest eigenvector, each measured at working precision sufficient to resolve ε_N above the floor. At every configuration tested, the natural eigenvector is essentially even and the natural and even-sector eigenvalues agree to hundreds of decimal digits.

λ^2	N	Precision	Evenness deviation	Eigenvalue rel. diff.	Verdict
13	120	HP-1000	2.126×10^{-966}	1.107×10^{-962}	essentially even
100	500	HP-1000	6.665×10^{-563}	2.444×10^{-559}	essentially even
1000	800	HP-2000	7.634×10^{-763}	1.559×10^{-758}	essentially even
1000	890	HP-2000	1.563×10^{-664}	1.807×10^{-660}	essentially even
1200	970	HP-2000	2.948×10^{-528}	7.927×10^{-526}	essentially even

Table 10: Computed ground-state separations at the precisions of Table 9. The last column is the even-state absolute residual divided by the computed same-sector separation. These HP values are numerical evidence, not analytic-matrix interval certificates.

λ^2	N	$o_0 - e_0$	$e_1 - e_0$	Residual / gap
13	120	3.05556×10^{-55}	1.31185×10^{-51}	1.13209×10^{-968}
100	500	1.79801×10^{-458}	1.78543×10^{-453}	2.70026×10^{-567}
1000	800	$1.24493 \times 10^{-1258}$	$2.05363 \times 10^{-1253}$	6.60626×10^{-768}
1000	890	$5.42414 \times 10^{-1357}$	$1.04167 \times 10^{-1351}$	1.55669×10^{-669}
1200	970	$2.86128 \times 10^{-1493}$	$6.55282 \times 10^{-1488}$	2.00727×10^{-533}

diagonal mass matrices, avoiding rounded normalization factors. Each bracket isolates exactly one sector eigenvalue, and $\sup I_{e_0} < \min\{\inf I_{e_1}, \inf I_{o_0}\}$ proves that the full ground state is simple and even. Table 11 reports the certified controls. A second implementation, using Schur updates, wider entry enclosures, and direct basis contractions, independently reproduces every inertia count at 500 decimal digits. These certificates concern the analytic finite matrices; a continuum extension also requires bounds on the omitted tail and coupling. They are separate verification supplements, not certificates inferred from Ultra capture receipts.

Table 11: Certified finite controls at $C = 13$. Displayed centers locate the exact rational brackets supplied with the verification supplement; each bracket has relative half-width 10^{-6} . Both full matrices are positive definite, with a simple even ground state.

N	e_0	o_0	e_1
10	2.83124×10^{-26}	1.64492×10^{-23}	5.49026×10^{-21}
120	3.48399×10^{-59}	3.05591×10^{-55}	1.31185×10^{-51}

The $\lambda^2=100$, $N=500$ row was remeasured with the current 64-bit guard reserve; the earlier 16-guard-bit calculation gave 5.102×10^{-548} . The additional 48 working bits supply approximately $48 \log_{10} 2 = 14.45$ decimal digits of arithmetic headroom, consistent with the roughly 15-order reduction to 6.665×10^{-563} . This change lowers the residual symmetry noise without changing the smallest eigenvalue, its sign, or the “essentially even” conclusion; the natural and even-sector eigenvalues differ by only 2.444×10^{-559} in relative terms.

The $\lambda^2=1000$ rows show the same guard-reserve effect. At $N=800$ the deviation falls from the earlier 2.941×10^{-748} to 7.634×10^{-763} , and at $N=890$ from 6.279×10^{-650} to 1.563×10^{-664} . Both improvements are approximately 14.6 decimal orders, consistent with the added 48 guard bits. The relative natural-versus-even eigenvalue differences are 1.559×10^{-758} and 1.807×10^{-660} , respectively, so the two routes remain numerically equivalent to hundreds of decimal digits.

The $\lambda^2=1200$, $N=970$ row was likewise remeasured with the 64-bit guard reserve. Its evenness deviation falls from 2.927×10^{-514} to 2.948×10^{-528} , an improvement of approximately 14.0 decimal orders. The natural and even-sector eigenvalues are numerically equivalent at the attainable relative precision, agreeing to approximately 525 decimal digits with relative difference 7.927×10^{-526} . Their full 6708-bit representations need not be bit-identical: the natural full-space solve and the reduced even-sector solve follow different high-precision operation sequences, so roundoff at the working-precision floor need not produce identical terminal bit patterns. Moreover, 6708 bits provide approximately 2019 decimal digits, while $|\varepsilon_N| \approx 6.799 \times 10^{-1499}$ consumes roughly 1499 orders of magnitude; the resulting ~ 520 relative-digit headroom is consistent with the observed 525-digit agreement. Thus the nonzero difference reflects the arithmetic floor rather than a mathematical discrepancy between the two eigenvalues.

At every configuration tested above the precision floor — including $\lambda^2 = 1200$, far beyond the range previously reported — the natural smallest eigenvector is even to hundreds of digits of precision, and the natural and even-sector solves converge to numerically indistinguishable eigenvalues at the reported accuracy. In the current $\lambda^2=13$, $N=120$, HP-1000 reproduction, performed at 3386-bit working precision, their relative difference is 1.107×10^{-962} . The unrestricted states and the two positive gaps provide empirical support for the finite simple-even hypothesis at these five configurations. We conjecture the same structure throughout the finite family: for every $\lambda > 1$ and $N \geq 1$, the lowest eigenspace of $QW_\lambda|_{E_N}$ is one-dimensional and even.

Proposition 1 (Residual control of parity). *Let $A = A^*$ commute with an orthogonal involution J , let $\|v\|_2 = 1$, let $\rho \in \mathbb{R}$, and set $r = Av - \rho v$. If ρ lies below the odd-sector spectrum by a distance at least $g_o > 0$, then*

$$\|v - Jv\|_2 \leq \frac{2\|r\|_2}{g_o}.$$

Proof. With $v_- = (I - J)v/2$, the odd projection gives $(A_- - \rho I)v_- = r_-$, where A_- is the odd restriction. Its inverse has norm at most g_o^{-1} , so $\|v - Jv\|_2 = 2\|v_-\|_2 \leq 2\|r\|_2/g_o$. $\square$

For the natural solver's stored infinity-relative residual $r_\infty = \|Av - \rho v\|_\infty/\|v\|_\infty$, a two-norm estimate is $\sqrt{2N+1} r_\infty \|v\|_\infty/\|v\|_2$. Together with the measured odd gaps, this gives parity upper estimates of 4.61×10^{-964} , 1.84×10^{-560} , 3.20×10^{-760} , 1.23×10^{-661} , and 8.14×10^{-526} in table order. Every measured parity defect lies below its corresponding estimate. This is a computed consistency check; certified residuals and gaps would turn the same calculation into a verified bound for the analytic matrix. If stored entries only approximately commute with reflection, the matrix-symmetry error must also enter the residual budget.

The proposition identifies the absolute gap, rather than the odd-to-even eigenvalue ratio alone, as the conditioning scale for parity. If $\|r\|_2 \leq K(\lambda, N)2^{-P_{\text{bits}}}$, then parity tolerance η follows from $P_{\text{bits}} \geq \log_2(2K/(\eta g_o))$. This is a conditional precision budget once K and g_o are bounded, and does not by itself give a root-accuracy budget.

Proposition 2 (Transfer to analytic finite roots). *Let q be an exact real unit approximate state and let the real symmetric matrix A have a simple ground state u . Suppose a real shift ρ lies below the remaining spectrum by at least $g > 0$. Put $R = \|Aq - \rho q\|_2$ and assume $R/g < 1$. After aligning the sign of u with q , one has*

$$\|u - q\|_2 \leq \eta := \sqrt{2} R/g.$$

For $H_q(t) = \sum_j q_j/(t - h_j)$, away from the poles,

$$|H_u(t) - H_q(t)| \leq \eta \left(\sum_j |t - h_j|^{-2} \right)^{1/2}, \quad |H'_u(t) - H'_q(t)| \leq \eta \left(\sum_j |t - h_j|^{-4} \right)^{1/2}.$$

Opposite endpoint signs and a derivative separated from zero after these error budgets are included certify a unique root of H_u .

Proof. Spectral decomposition bounds the component of q orthogonal to u by R/g . If its norm is $\sin \theta$, then the aligned unit vectors satisfy $\|u - q\|_2 \leq \sqrt{2} \sin \theta$. The two function bounds follow from Cauchy–Schwarz. Continuity and the derivative bound give existence and uniqueness of the root. For an exactly even q , the same argument can use the remaining even-sector spectrum. $\square$

At $C = 13$, the analytic interval matrices give $\eta < 3.83 \times 10^{-130}$ for $N = 10$ and $\eta < 2.23 \times 10^{-242}$ for $N = 120$. With these errors included, all ten $N = 10$ brackets and the $N = 120$ bracket near γ_1 pass the endpoint and derivative tests. The first five $N = 10$ comparisons retain the matching depths in Table 3; the $N = 120$ comparison has $D_1 \in [55.7635578784, 55.7635578796]$. For $N = 10$, ten distinct positive roots exhaust a numerator of degree at most ten, so the complete count and the stated full-spectrum ordinals transfer to the analytic finite source. An independent calculation using wider matrix enclosures, an infinity-norm residual bound, and the smaller full-space spectral gap verifies all eleven brackets. These are finite CCM results, without taking a continuum or increasing-cutoff limit.

Fixed-matrix precision control. We round one fixed $C = 13$, $N = 10$ matrix to 24, 53, 64, 80, 96, 128, 192, 256, and 384 bits, then solve each rounded problem at 220 decimal digits. This separates matrix-entry rounding from assembly and solver precision. At 24 bits the lowest state is odd, so the even-source root comparison is inapplicable. At 53 and 64 bits the lowest eigenvalue is negative, with first-zero matching depths of 16.971 and 17.759 digits. At 192 bits and above the positive ground value and 21.585231815758-digit match are recovered. Thus entry rounding alone can produce the apparent failure and its recovery; the threshold is specific to the configuration.

The precision-floor artifact. When $|\log_{10} \varepsilon_N|$ approaches or exceeds the working precision in digits, the eigenvalue sinks below the arithmetic’s resolving power. In this regime the computed eigenvector is under-resolved and becomes sensitive to rounding within numerically indistinguishable directions. The result can *appear* non-even (deviation $\sim O(1)$), while the eigenvalues can *appear* to split between the natural and even-sector paths. Under the earlier guard policy, the HP-1000 $\lambda^2 = 1000$ configuration ($|\log_{10} \varepsilon_N| \approx 1264$, well past the available arithmetic floor) exhibits this artifact: deviation 1.87, eigenvalue split 24%. Increasing the requested precision to HP-2000 under the same policy at the *identical* ($\lambda^2=1000, N=800$) configuration collapses the apparent breakdown entirely: deviation 2.94×10^{-748} , with eigenvalues numerically equivalent at the reported precision. Thus the only variable in that controlled comparison is requested precision. The latest 64-guard-bit HP-2000 run uses 6708 working bits (approximately 2019.3 decimal digits, a floor ratio of approximately 1.60) and reduces the residual symmetry noise further to 7.634×10^{-763} , again with numerically equivalent natural and even-sector eigenvalues at the reported precision.

The even-sector restriction is empirically unnecessary. To further test whether the eigenvector is naturally even, we ran the full CCM pipeline (eigenstate solve $\rightarrow$ eigenvector $\rightarrow R(s)$ zero-finding $\rightarrow$ Riemann-zero matching digits) with the unrestricted natural full-space policy at configurations spanning $\lambda^2 = 13$ –1000, $N = 10$ –800, and HP-200 through HP-2000. At every tested above-floor point, the natural path produces eigenvalues and Riemann-zero approximations numerically indistinguishable from the reduced even-sector path at the reported accuracy. The direct restriction is therefore not needed to obtain the reported values at these points, although it remains the faster paper-reproduction default.

5 Discussion

5.1 Where the Construction’s Power Lies

The CCM construction’s measured accuracy is controlled jointly by three caller-selected parameters:

- λ sets the arithmetic cutoff and changes the intrinsic finite-configuration depth;
- N controls progress along the finite-basis ramp toward that depth;
- working precision determines how much of the finite result can be resolved without floor artifacts.

Along the sampled path $N/\sqrt{\lambda^2} \approx 25\text{--}30$, the HP-2000 runs at $(\lambda^2, N) = (1000, 890)$ and $(1200, 970)$ achieve 1358.79 and 1495.14 first-root matching digits, respectively. These are achieved finite accuracies, distinct from the requested precision. The ratio is not a universal budget for reaching the finite-cutoff plateau: Tables 5 and 7 still show substantial gains from additional modes beyond that ratio. A companion convergence-law study [9] examines the broader coordinate $N/(\lambda^2 \ln \lambda^2)$; the present reproduction paper does not assume that model. In practice, λ , N , and precision should be chosen together for the desired accuracy target.

5.2 The Prime-Weight Profile

Each prime power $k = p^j$ contributes weight $\Lambda(k)/\sqrt{k} = \log(p)/\sqrt{k}$ to the Weil form. For primes ($j = 1$), this is $\log(p)/\sqrt{p}$, whose derivative vanishes at $p = e^2 \approx 7.3891$ with maximum value $2/e \approx 0.73576$.

Table 12: Individual prime contributions $\log(p)/\sqrt{p}$ for the primes captured by $\lambda^2 = 13$. The maximum value $2/e$ is attained at the (non-prime) point $p = e^2 \approx 7.3891$; the integer prime 7 achieves 99.96% of this maximum.

p	$\log(p)/\sqrt{p}$	% of $2/e$
2	0.49013	66.62%
3	0.63428	86.21%
5	0.71976	97.83%
7	0.73548	99.96%
11	0.72299	98.26%
13	0.71139	96.69%
$\max = 2/e \text{ at } p = e^2$		100%

The following observation describes a property of the resulting finite prime set; it does not imply that the original cutoff was selected for this reason.

The CCM paper’s choice of $\lambda^2 = 13$ has the useful consequence that it captures the peak prime $p = 7$ to within 0.04% of the maximum value and includes both adjacent primes ($p = 5$ at 97.8% of the maximum and $p = 11$ at 98.3%) that lie within $\sim 2\%$ of the peak. The choice covers the entire near-peak cluster of primes.

Past the peak, $\log(p)/\sqrt{p}$ decays as a power law with a logarithmic factor (0.46052 at $p = 100$, 0.21844 at $p = 1000$, 0.092103 at $p = 10000$). Thus later primes still have visible individual weights at the finite cutoffs studied here. Their cumulative effect is consistent with the measured reduction in $|\varepsilon_N|$, but the scalar weight profile alone does not derive the eigenvalue-decay law; matrix interactions and the simultaneous scaling of N remain essential.

Measured prime-power responses. At $C = 13$, $N = 120$, we independently test each retained prime-power matrix direction D_q by solving the perturbed family $A + tD_q$ with fixed observation poles. All nine directions pass at $t = 10^{-90}, 10^{-100}, 10^{-110}$. Centered-difference errors decrease quadratically; the final relative discrepancies in first-root velocity are below 1.1×10^{-117} . Powers 2, 3, 5, and 7 contribute approximately -226.280 , -16.1755 , -3.15035×10^{-4} , and -1.71768×10^{-10} , respectively. The scalar weight maximum at 7 therefore does not identify the largest local root response.

The source-matched retained decomposition contains approximately $+274.545$ from the pole-correction matrix, -26.5174 from the archimedean matrix, -242.517 from the aggregate prime matrix, and -5.51072 from moving secular poles. Their full-precision sum is -4.61362×10^{-53} ; the rounded displayed components cannot reproduce that cancellation. The aggregate is reconciled from retained measurements, rather than an independent validation of every nonprime derivative. A separate one-sided matrix check also verifies the prime-13 activation derivative. These results identify local response and cancellation, not finite prime-removal effects or an asymptotic convergence law.

5.3 Implications for the Proof Strategy

Our findings have implications for the CCM proof strategy:

1. The unrestricted evenness measurements and both spectral gaps (Tables 9 and 10) support a simple even finite ground state through $\lambda^2 = 1200$ at HP-2000. The corresponding conjecture has a concrete certification target: strict separation of the first even value from the next even and lowest odd values. The two finite instances in Table 11 now meet that target rigorously. Proposition 1 connects observed parity noise to the residual and absolute odd-sector gap.
2. The basis sweep in Table 2 shows decreasing gains across the ramp: the first three increments yield approximately 1.421, 0.871, and 0.579 first-root digits per added mode. Gains become small near the observed plateau. At larger cutoffs, Tables 5 and 7 show that substantial improvements from increasing N remain possible.
3. The rapid measured change of ε_N along the sampled (λ, N) paths supplies empirical convergence evidence, but both a rigorous two-parameter law and the asymptotic decay class remain open.

6 Conclusion

We have independently reproduced and significantly extended the CCM zeta spectral triple construction, achieving 1495.14 measured matching digits on the first Riemann zero at $(C, N) = (1200, 970)$ with 6708 working bits. The 1019.0-to-1260.36-digit paired-precision experiment at $(C, N) = (1000, 800)$ exposes the transition from arithmetic limitation to resolved finite-configuration error. The key findings are:

- A smoothly saturating basis-size ramp that is approximately linear over the observed early range.
- Rapid empirical growth of the Weil-eigenvalue depth along the tested cutoff-and-basis paths, without a claim that prime count alone determines the rate.
- Observed monotone growth of first-zero accuracy across the sampled fixed- N cutoff sweep, with a slowly varying conversion gap between root accuracy and $-\log_{10} |\varepsilon_N|$.

- Critical- N accuracy thresholds resolved on integer brackets: at $C = 13$, the first zero crosses 50 digits at $N = 40$ and 55 digits at $N = 56$, with agreement between HP-200 and HP-1000.
- Numerical support for a simple even finite ground state at the five headline parity configurations through $\lambda^2 = 1200$, with a residual-to-gap explanation for parity noise, and rigorous finite matrix and root certificates at $C = 13$, $N = 10$ and $N = 120$.
- Independent integral-assembly, fixed-matrix precision, and prime-response controls supporting the numerical mechanisms.
- An exact reference-free census for the serialized 21×21 example, preserving its beyond-band matches and specifying their full positive spectral positions.
- Notable low-dimensional compression: a 21×21 matrix from six primes already produces 21.585 matching digits.

Together these results provide a comprehensive empirical foundation for future theoretical work on the construction.

The observed compression suggests algebraic structure in the Weil quadratic form that is not yet theoretically understood. Explaining why this mechanism produces Riemann zeros with such precision remains an open problem whose resolution could advance the spectral approach to the Riemann Hypothesis.

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Data and Code Availability

The complete paper-specific source code and reproduction scripts are available at:

<https://github.com/TeamXcelerator/ccm-reproduction-and-convergence>

The v2.4 tables originated with Toolkit 0.14.2; the completed September 2026 v2.5 campaign assessed all 14 individual claim scripts over 53 journals using the Toolkit 0.15.0 pin recorded in Cargo.lock. Numerical and publication summaries passed with the documented Claim 1c and

Claim 8 applicability qualifications. The high-precision first-root column, parity-gap table, and exact small-source census augment the historical measurements. Source identities distinguish reused artifacts from fresh computation and bind the recovered high-precision diagnostics. Supporting code and aggregate evidence are supplied in the repository. The reproduction scripts reuse compatible cached artifacts when available and compute locally otherwise.

The shared mathematical library and content-bound reference-zero dataset are supplied by the Xcelerator Toolkit at:

<https://github.com/TeamXcelerator/xcelerator-toolkit>

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